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Senior Project 1

Smart Traffic Signal

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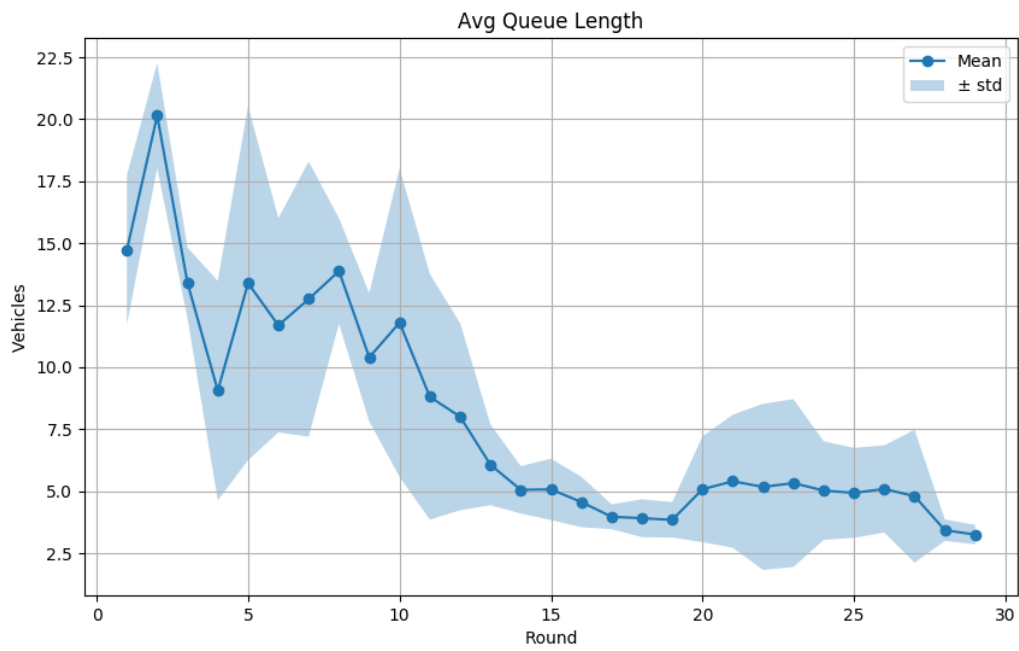


Figure 1: AVG Queue Length

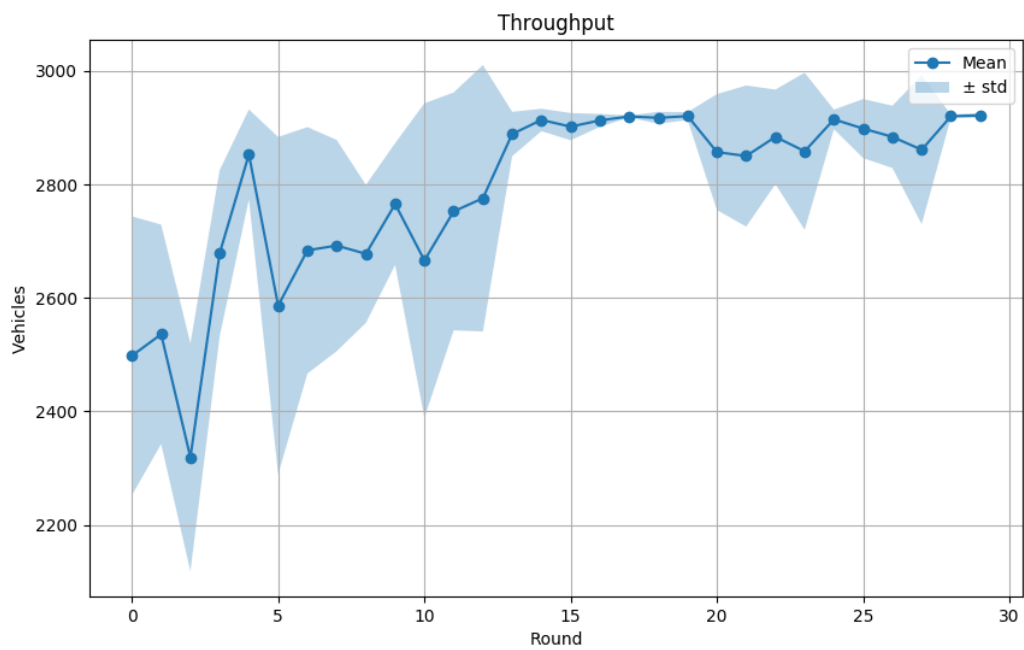


Figure 2: Throughput

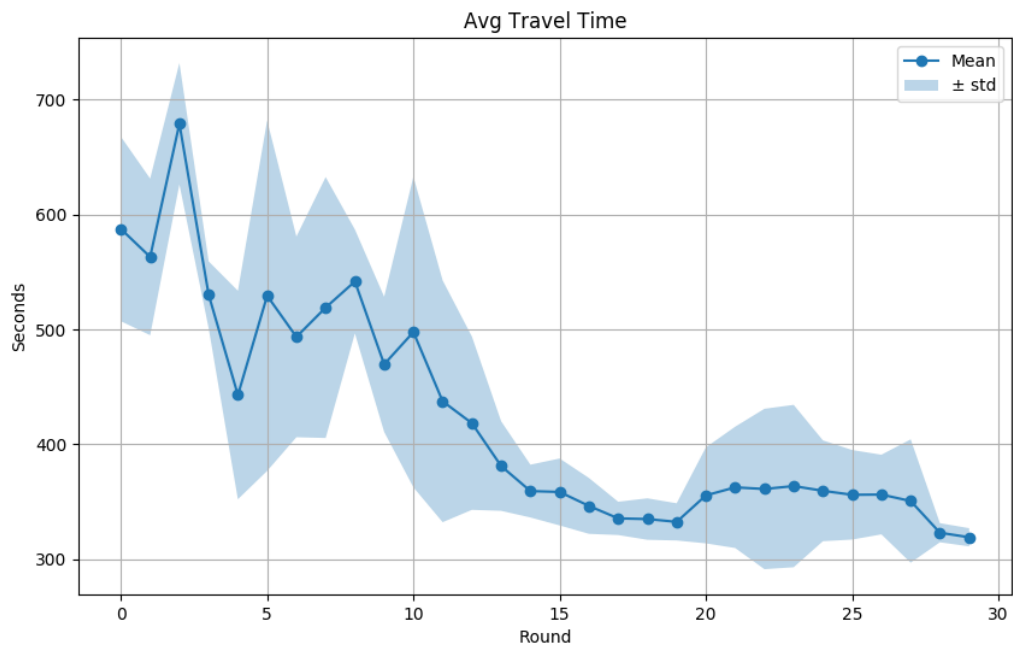


Figure 3: AVG Travel Time

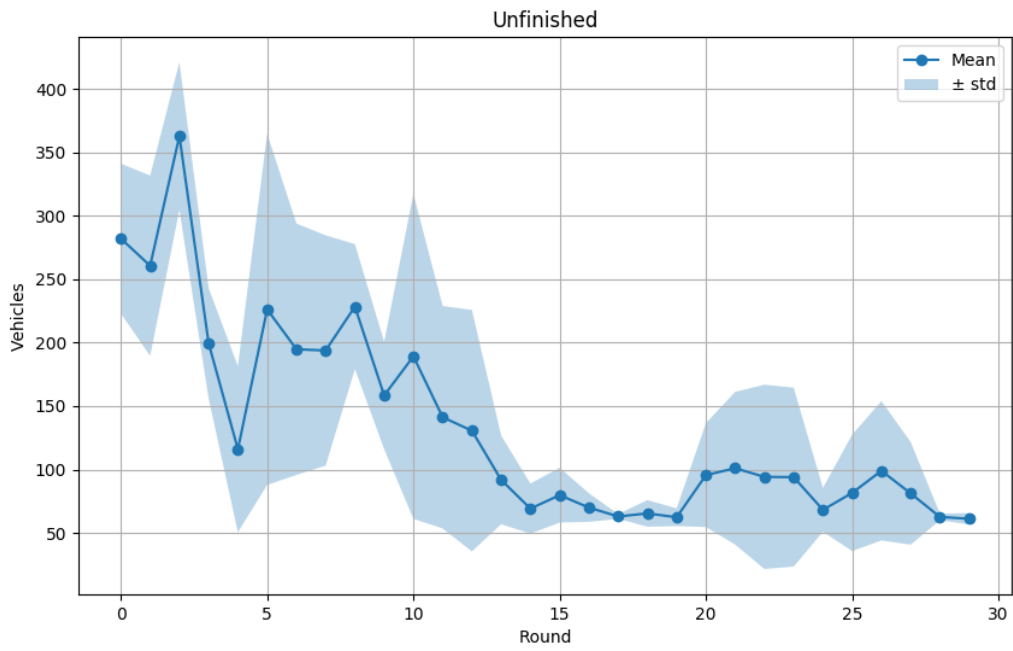


Figure 4: Unfinished Vehicles

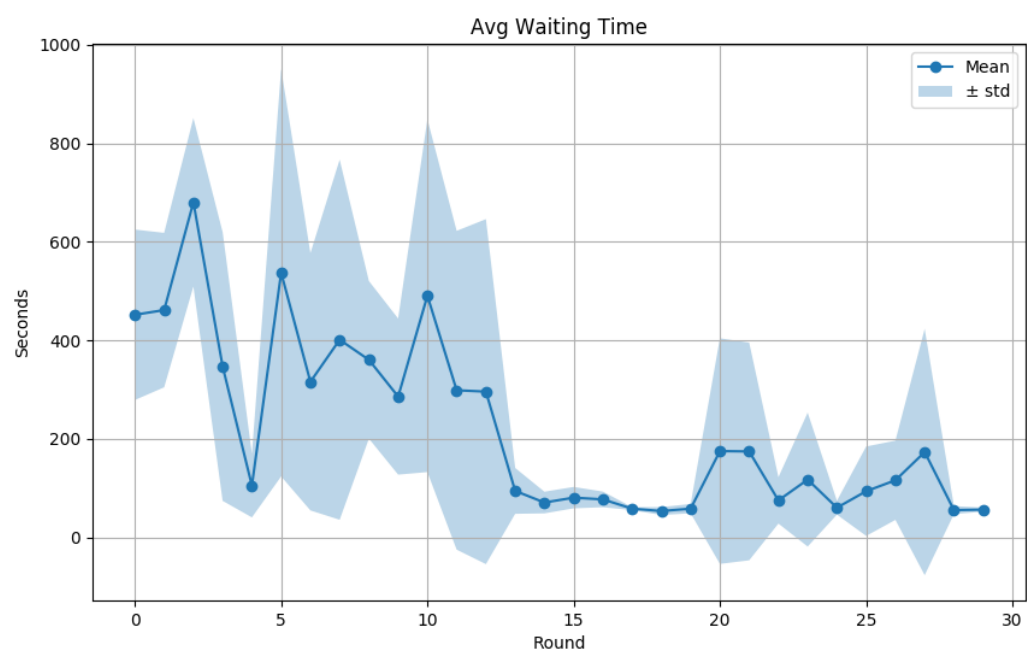


Figure 5: AVG Waiting Time

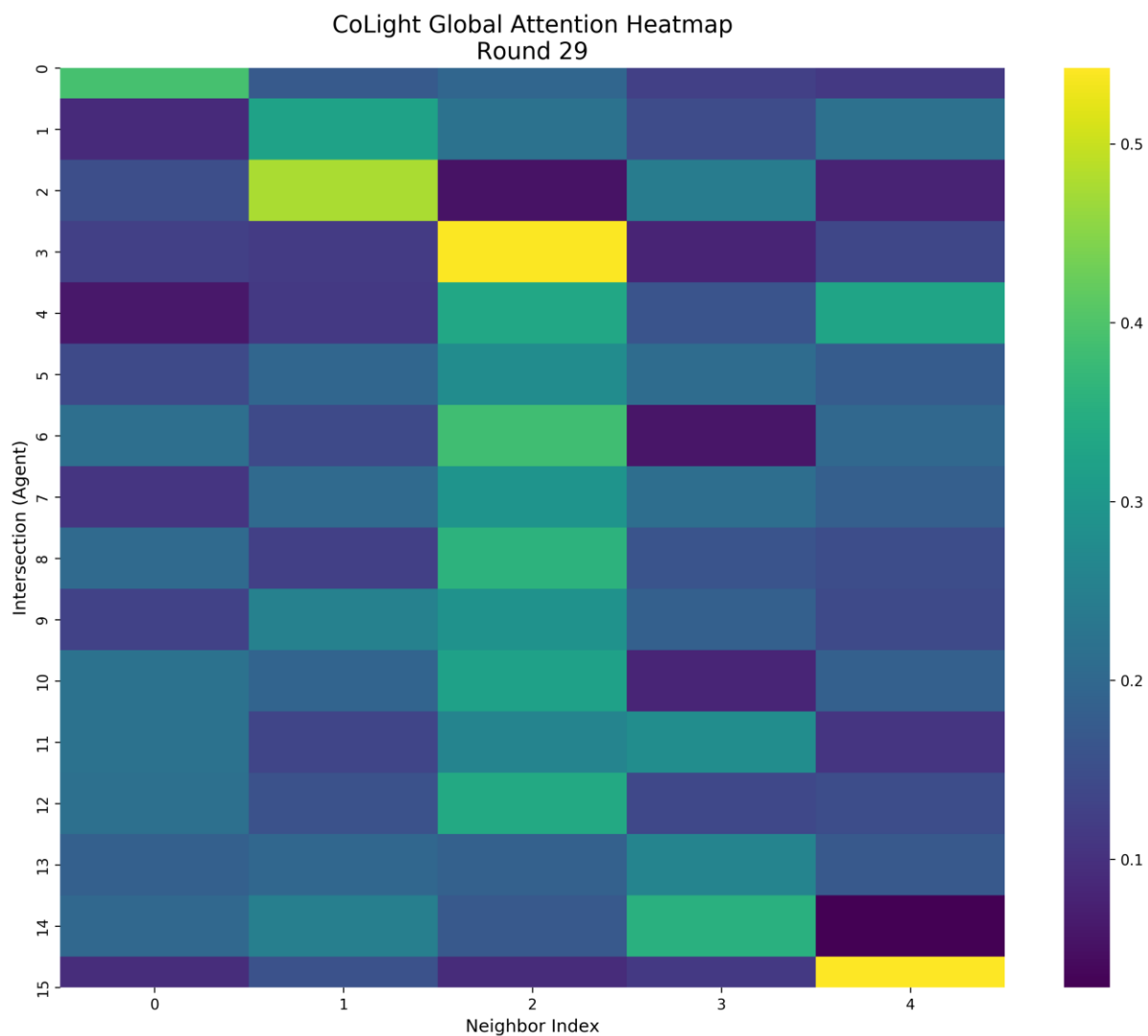


Figure 6: Global Attention Heatmap for exp-1

List of Equations

1. Queue Length (Average Queue Length)

For each sample(s) in a pickle file:

$$\left[Q_s = \sum_{l=1}^L N_{l,s} \right]$$

where:

- ($N_{l,s}$)= number of vehicles stopped (thresholded) on lane (l)
- (L) = number of lanes

Average over samples:

$$\left[\bar{Q} = \frac{1}{S} \sum_{s=1}^S Q_s \right]$$

Final average over all pickle files:

$$\left[Q_{\text{avg}} = \frac{1}{K} \sum_{k=1}^K \bar{Q}_k \right]$$

2. Vehicle Travel Time

For each record:

$$\left[T_i = t_i^{\{\text{leave}-t_i^{\{\text{enter}\}}\}} \right]$$

If a vehicle has multiple segments:

$$[T_v = \sum_{i \in v} (t_i^{\{\text{leave}\}} - t_i^{\{\text{enter}\}})]$$

Average travel time:

$$[\bar{T} = \frac{1}{|V|} \sum_{v \in V} T_v]$$

where:

- (V) = set of all vehicles

3. Throughput (Completed Vehicles)

$$[\text{Throughput} = |V_{\text{completed}}|]$$

where:

$$[V_{\text{completed}} = v \in V \mid t_v^{\text{leave}} \neq \text{NaN}]$$

4. Total Vehicles

$$[|V| = \text{number of unique vehicle IDs}]$$

5. Unfinished Vehicles

$$[V_{\text{unfinished}} = |V| - |V_{\text{completed}}|]$$

6. Average Waiting Time (Unfinished Vehicles Only)

For each unfinished vehicle (v):

$$[W_v = T_{\text{end}} - t_v^{\text{enter}}]$$

where:

- $(T_{\text{end}} = 3600) \text{ seconds}(\text{simulationhorizon})$

Average waiting time:

$$\begin{aligned} [W] &= \frac{1}{|V_{\text{unfinished}}|} \\ &\sum_{v \in V_{\text{unfinished}}} (T_{\text{end}} - t_v^{\text{enter}}) \end{aligned}$$

7. Simulation Horizon Constant

$$[T_{\text{end}} = 3600]$$

List of Terms and Symbols

Symbol / Term	Meaning / Description
(S)	State space in Reinforcement Learning
(A)	Action space in Reinforcement Learning
(R)	Reward signal received from the environment
(G_t)	Cumulative discounted reward (return) at time step (t)
(γ)	Discount factor used in Reinforcement Learning
$(r_{\{t+k+1\}})$	Future reward received at step $(t + k + 1)$
(Q_s)	Queue length value for a single traffic sample
$(\bar{Q})$	Average queue length over all samples
(Q_{avg})	Final average queue length across all pickle files
$(N_{l,s})$	Number of stopped vehicles on lane (l) in sample (s)
(L)	Total number of lanes
$(S)(QueueEquation)$	Number of traffic samples
(K)	Number of processed pickle files
(T_i)	Travel time for a vehicle segment
(t_i^{leave})	Vehicle leave time
(t_i^{enter})	Vehicles enter time
(T_v)	Total travel time for vehicle (v)
$(\bar{T})$	Average travel time
(V)	Set of all vehicles
$(V_{completed})$	Set of completed vehicles
$(V_{unfinished})$	Set of unfinished vehicles
$(Throughput)$	Number of completed vehicles
(W_v)	Waiting time for unfinished vehicle (v)
$(\bar{W})$	Average waiting time
(T_{end})	Simulation horizon duration
ATT	Average Travel Time
AWT	Average Waiting Time
AQL	Average Queue Length
RL	Reinforcement Learning
DRL	Deep Reinforcement Learning
MARL	Multi-Agent Reinforcement Learning
DQN	Deep Q-Network
GAT	Graph Attention Network
MAS	Multi-Agent System
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
V2I	Vehicle-to-Infrastructure Communication

SUMO	Simulation of Urban Mobility traffic simulator
CoLight	Graph-based multi-agent RL traffic signal control framework
PressLight	Pressure-based reinforcement learning traffic signal controller
MPLight	Multi-Agent PressLight traffic control framework
GPLight	Group-based reinforcement learning traffic signal controller
CVLight	Connected Vehicle-based traffic signal control framework
MOMA-DDPG	Multi-objective Multi-Agent Deep Deterministic Policy Gradient
roadnet.json	File describing road network topology
flow.json	File defining vehicle generation and traffic demand
Experience Replay	RL mechanism for storing and reusing experiences during training
Target Network	Stabilization network used in DQN training
Epsilon-Greedy	Exploration strategy balancing exploration and exploitation
Queue Length	Number of stopped vehicles waiting at intersections
Throughput	Total number of vehicles successfully completed
Unfinished Vehicles	Vehicles that did not complete their routes during simulation
State Representation	Encoded traffic features observed by the agent
Reward Function	Function defining optimization objectives for the agent
Graph Representation	Spatial representation of intersections and connections
Training Round	One iteration cycle of reinforcement learning training
Random Seed	Fixed initialization value ensuring reproducibility
Standard Deviation (Std)	Statistical measure of result variability
Mean	Average value across multiple experiments
Hangzhou 1×1	Single-intersection traffic dataset
Hangzhou 4×4	Multi-intersection traffic dataset with 16 intersections
Fixed-Time Control	Traditional traffic signal timing method using predefined schedules
Actuated Control	Sensor-based adaptive traffic signal control
Green Wave	Coordinated signal timing strategy for continuous vehicle flow
Temporal Difference (TD) Loss	Loss used to update predicted Q-values
MSE	Mean Squared Error
Policy	Strategy used by the RL agent to select actions
Environment	Traffic simulation system interacting with the agent
Agent	Decision-making entity controlling traffic signals

Abstract

This study presents the design and implementation of a smart traffic signal system aimed at improving urban traffic efficiency and reducing congestion across multiple interconnected intersections. Traditional traffic control systems operate on fixed timing schedules, which often fail to adapt effectively to dynamic and real-time traffic variations. The proposed system utilizes sensors, cameras, and data-driven techniques to dynamically adjust signal timings based on current traffic density and flow conditions within a network of intersections. A reinforcement learning-based approach is adopted, enabling the system to learn optimal traffic signal control policies through continuous interaction with the traffic environment. The model observes real-time state variables, such as vehicle queue lengths, waiting times, and lane occupancy, across multiple intersections and selects coordinated actions that maximize overall traffic flow efficiency. By considering the interdependence between adjacent intersections, the system learns to reduce bottlenecks and prevent congestion of propagation. Through iterative learning, the model refines its control strategy by optimizing cumulative rewards associated with minimizing delays and improving throughput across the network.

Simulation results demonstrate that the proposed multi-intersection control system significantly outperforms conventional fixed-time and isolated signal control methods. Notable improvements are observed in average delay reduction, queue length minimization, and smoother traffic progression along corridors. The system maintains robust performance under varying traffic conditions, including peak hours and uneven traffic distributions. The architecture is scalable and designed to support coordinated control over large urban areas, making it suitable for real-world deployment. Its modular structure allows for seamless expansion and adaptation to different traffic environments. Overall, the proposed reinforcement learning-based smart traffic signal system provides an efficient and adaptive solution for managing complex urban traffic networks and contributes to the advancement of intelligent transportation systems.

تمهيد

تقدم هذه الدراسة تصميم وتنفيذ نظام إشارات مرور ذكي يهدف إلى تحسين كفاءة حركة المرور الحضرية وتقليل الازدحام عبر عدة تقاطعات مترابطة. تعتمد أنظمة التحكم المروري التقليدية على جداول زمنية ثابتة، والتي غالباً ما تفشل في التكيف مع التغيرات الديناميكية والفعالية في حركة المرور. يستخدم النظام المقترح أجهزة استشعار وكاميرات وتقنيات قائمة على البيانات لضبط توقيت الإشارات بشكل ديناميكي بناءً على كثافة وتدفق حركة المرور الحالية ضمن شبكة من التقاطعات. تم اعتماد نهج قائم على التعلم المعزز، مما يمكن النظام من تعلم سياسات تحكم مثلى لإشارات المرور من خلال التفاعل المستمر مع بيئة المرور.

يقوم النموذج بمراقبة متغيرات الحالة في الزمن الحقيقي، مثل أطوال طوابير المركبات، وأزمنة الانتظار، وإشغال المسارات عبر عدة تقاطعات، ثم يختار إجراءات منسقة تهدف إلى تعظيم كفاءة تدفق المرور بشكل عام. ومن خلال مراعاة الترابط بين التقاطعات المجاورة، يتعلم النظام تقليل الاختناقات ومنع انتشار الازدحام. ومع التكرار المستمر، يقوم النموذج بتحسين استراتيجيته في التحكم من خلال تعظيم العوائد التراكمية المرتبطة بتقليل التأخير وتحسين الانسيابية عبر الشبكة.

تُظهر نتائج المحاكاة أن نظام التحكم المقترح متعدد التقاطعات يتفوق بشكل ملحوظ على الأنظمة التقليدية ذات التوقيت الثابت أو التحكم المنفصل لكل تقاطع. حيث لوحظت تحسينات كبيرة في تقليل متوسط زمن التأخير، وتقليص أطوال الطوابير، وتحقيق انسيابية أفضل لحركة المرور على طول المحاور. كما يحافظ النظام على أداء مستقر في ظل ظروف مرورية مختلفة، بما في ذلك أوقات الذروة والتوزيعات غير المتوازنة لحركة المرور.

تم تصميم البنية المعمارية للنظام بحيث تكون قابلة للتوسع لدعم التحكم المنسق عبر مناطق حضرية واسعة، مما يجعلها مناسبة للتطبيق العملي. كما أن تصميمها المعياري يسمح بالتوسع والتكيف بسهولة مع بيئات مرورية مختلفة. بشكل عام، يوفر نظام إشارات المرور الذكي المعتمد على التعلم المعزز حلاً فعالاً وقابلاً للتكيف لإدارة شبكات المرور الحضرية المعقدة، ويساهم في تطوير أنظمة النقل الذكية.

Chapter one: Introduction

1.1 Introduction

Urban traffic congestion remains one of the most pressing challenges in modern cities, leading to increased travel time, fuel consumption, and environmental pollution.

Conventional traffic signal control systems rely on fixed or semi-adaptive timing strategies, which are often insufficient for handling the dynamic and stochastic nature of real-world traffic flows. With the rapid growth of urbanization and vehicle usage, there is a critical need for intelligent and adaptive traffic management solutions that can respond effectively to varying traffic conditions.

This project proposes a reinforcement learning (RL)-based smart traffic signal control system designed for coordinated management of multiple interconnected intersections. Unlike traditional approaches, the proposed system models traffic signal control as a sequential decision-making problem, where each intersection operates as an agent that learns optimal policies through continuous interaction with its environment. The framework emphasizes coordination among intersections to reduce congestion propagation and improve overall network performance.

The methodology is centered on a multi-agent reinforcement learning approach, where agents observe the traffic state in real time and make decisions regarding signal phase selection. A key contribution to this work lies in the modification of state representation, where relevant traffic features such as queue lengths, waiting times, and lane-level conditions are encoded more effectively to improve learning efficiency. Additionally, the reward function is carefully redesigned to better capture system-level objectives, including minimizing cumulative delays and enhancing traffic throughput across the network.

The model is trained and evaluated within a simulated traffic environment that replicates realistic multi-intersection scenarios. Through iterative learning, agents develop cooperative strategies that balance local intersection performance with global traffic efficiency. The results demonstrate that the proposed approach can significantly improve traffic flow compared to conventional methods. Overall, this work highlights the potential of reinforcement learning in developing scalable and adaptive traffic signal control systems for complex urban networks.

1.2 The Scientific Problem

The scientific problem addressed in this project is the inefficiency of traditional traffic signal control systems in handling dynamic and complex urban traffic conditions. Fixed-time and actuated control methods lack the adaptability required for real-time traffic fluctuations, especially in networks of multiple interconnected intersections. This leads to increased congestion, longer delays, and inefficient traffic flow. The challenge becomes more significant when considering the interdependence between adjacent intersections, where local decisions can negatively impact global performance. Existing approaches often fail to achieve coordinated optimization across the entire network. Furthermore, representing traffic states effectively and designing appropriate reward functions remain critical challenges in reinforcement learning-based control. Poor state representation can limit the model's understanding of traffic conditions, while inadequate reward design can lead to suboptimal policies. Therefore, the core problem is to develop an adaptive, scalable, and coordinated traffic signal control strategy that can learn efficient policies for multi-intersection environments. This includes improving state representation and reward formulation to enhance learning performance and overall traffic efficiency.

1.3 Objectives

The main objectives of this research are as follows:

The primary objective of this research is to enhance the original CoLight reinforcement learning model for intelligent traffic signal control by improving the reward function, state representation, and clipping factor to achieve better traffic flow optimization and congestion reduction across multiple interconnected intersections. The proposed system is designed to adapt dynamically to real-time traffic conditions and optimize signal timing decisions efficiently.

Another objective is to evaluate the effectiveness and stability of the enhanced CoLight model through a rigorous experimental setup. The evaluation was conducted using the real-world Hangzhou traffic dataset, ensuring accurate and reliable performance analysis.

To improve the validity of the experimental results, five different random seed values were used. The model was trained through five separate experiments, where each experiment utilized a different seed value and was trained for 30 rounds. The mean and standard deviation of the results were then computed to provide a comprehensive evaluation of the model's performance and consistency.

In addition, the research aims to compare the enhanced CoLight model directly with the original CoLight implementation without modifications. Both models were evaluated under the same conditions using the same five random seeds and experimental settings to ensure a fair and accurate comparison of performance improvements.

1.4 Research Contribution

The main contributions of this research can be summarized as follows:

- Enhancing the original CoLight reinforcement learning model through modifications to the state representation, reward function, and clipping factor to improve intelligent traffic signal control performance.
- Developing an improved state representation capable of capturing real-time traffic conditions more effectively, allowing better decision-making across multiple interconnected intersections.
- Designing a modified reward function that supports traffic optimization objectives, including reducing congestion, minimizing vehicle delay, and decreasing queue lengths within the traffic network.
- Improving the training stability and learning efficiency of the reinforcement learning process by adjusting the clipping factor within the model.
- Conducting a comprehensive evaluation using the real-world Hangzhou traffic dataset within a realistic traffic simulation environment.
- Performing rigorous experiments using five different random seed values, where five separate experiments were trained for 30 rounds to ensure reliable and unbiased evaluation results.
- Computing the mean and standard deviation of the experimental results to measure the consistency, robustness, and overall effectiveness of the proposed enhancements.
- Comparing the enhanced CoLight model directly with the original CoLight implementation under the same experimental settings and random seed configurations to ensure a fair and accurate performance comparison.
- Demonstrating improvements in important traffic performance metrics, including traffic delay, queue length, and overall traffic flow efficiency under varying traffic conditions.

1.5 Research beneficiaries

The proposed research can benefit several sectors and stakeholders involved in transportation management and intelligent traffic systems. The main beneficiaries of this work are as follows:

- Traffic management authorities can benefit from smart traffic signal control by improving traffic flow efficiency and reducing congestion at multiple interconnected intersections through adaptive signal timing decisions.
- Municipal governments and urban planners can utilize the proposed system to support the planning and management of urban transportation infrastructure using data-driven and intelligent traffic control strategies.

- Transportation agencies can apply the enhanced reinforcement learning model to optimize traffic signal coordination across large-scale road networks and improve overall intersection performance.
- Drivers and daily commuters can benefit from reduced vehicle waiting times, shorter travel durations, and smoother traffic movement resulting from improved traffic signal control.
- Environmental and sustainability sectors can benefit from reduced fuel consumption and lower vehicle emissions due to decreased traffic congestion and idle waiting times at intersections.
- Researchers and academic institutions can use this work as a reference for further studies in reinforcement learning, multi-agent systems, and intelligent transportation systems, particularly in improving state representation and reward design within traffic signal control models.
- Smart city projects and intelligent transportation initiatives can benefit from integrating the proposed enhanced CoLight framework into advanced urban mobility systems to support more efficient and adaptive traffic management solutions.
- Developers and engineers working in intelligent transportation technologies can benefit from the experimental evaluation methodology used in this research, including the use of five random seed values, repeated experiments, and statistical analysis through mean and standard deviation calculations for reliable performance assessment.

1.6 Thesis Organization

This thesis is organized into six chapters in addition to the references section, as follows:

- **Chapter One: Abstract**
- This chapter presents an overview of the research, including the introduction, scientific problem, objectives, research contributions, research beneficiaries, and thesis organization.
- **Chapter Two: Dataset**

This chapter describes the datasets used in the research, including the Hangzhou 1×1 and Hangzhou 4×4 traffic datasets. It also explains the structure of the datasets, including roadnet.json and flow.json, and discusses their role in training and evaluating the reinforcement learning model.

- **Chapter Three: Literature Review**

This chapter reviews previous studies and related work in intelligent transportation systems, reinforcement learning, multi-agent traffic signal control, and graph-based traffic optimization methods. It also includes comparisons between existing approaches such as CoLight, PressLight, GPLight, and other reinforcement learning-based traffic control systems.

- **Chapter Four: Research Methodology**

This chapter explains the proposed methodology and the architecture of the enhanced CoLight model. It presents the modifications applied to the state representation, reward

function, graph representation, clipping factor, and training configuration. In addition, it describes the reinforcement learning pipeline, simulation environment, training algorithms, and evaluation procedures.

- **Chapter Five: Research Results**

This chapter presents the experimental results obtained from the enhanced CoLight model. It includes performance evaluation using five different random seed values, statistical analysis using mean and standard deviation, and comparisons with the original CoLight implementation under the same experimental settings.

- **Chapter Six: Conclusion**

This chapter summarizes the research findings, contributions, and overall outcomes of the proposed work. It also discusses the effectiveness of the enhanced reinforcement learning model in improving traffic signal control performance.

- **References**

This section contains all references, research papers, datasets, and academic resources used throughout the thesis.

Chapter two: Dataset

Traffic signal control has been extensively studied in the field of intelligent transportation systems. Traditional approaches struggle to handle dynamic and complex urban traffic conditions. Recent developments focus on adaptive and learning-based methods, especially reinforcement learning. Multi-intersection coordination has emerged as a critical research direction. This section reviews related work and identifies existing research gaps.

2.1 Introduction

The rapid growth of urban populations and vehicle usage has led to increasing challenges in traffic management systems worldwide. Conventional traffic signal control methods, such as fixed-time and actuated systems, are often unable to respond effectively to real-time variations in traffic demand. As a result, these methods frequently lead to congestion, increased travel delays, and inefficient road utilization.

In recent years, researchers have explored intelligent and data-driven approaches to overcome these limitations. Among these, reinforcement learning has gained significant attention due to its ability to learn optimal control policies through interaction with dynamic environments. This approach is particularly suitable for traffic signal control, where system conditions change continuously and unpredictably.

Furthermore, the coordination of multiple intersections has become an important research focus, as isolated control strategies often fail to optimize overall network performance. Therefore, this literature review examines existing studies on both single and multi-intersection traffic control systems, with an emphasis on reinforcement learning-based approaches and their effectiveness in improving urban traffic efficiency.

2.2 Dataset Overview

In our research, we work on the Hangzhou 4×4 traffic dataset, which is a benchmark dataset for multi-intersection traffic signal control using reinforcement learning. The dataset is mainly composed of two core files: **roadnet.json** and **flow.json**, which together define the traffic network structure and traffic demand.

The **roadnet.json** file describes the complete road network layout. It includes the definition of intersections, road segments, lanes, and connectivity between different nodes in the 4×4 grid structure. Each intersection is represented with its geometric information, signal configuration, and lane connections. It also specifies how incoming and outgoing lanes are linked, allowing vehicles to move realistically between intersections. This file essentially defines the spatial structure of the traffic environment and is crucial for building the simulation topology.

The **flow.json** file defines the traffic demand within the network. It contains information about vehicle generation, including the origin and destination of vehicles, entry rates, routes, and timing of vehicle injections into the system. It models dynamic traffic patterns by specifying how many vehicles are generated at different time intervals and how they traverse the network. This file is responsible for simulating realistic traffic flow conditions such as peak and off-peak demand.

Together, these two files form a complete simulation environment: **roadnet.json** defines the physical road infrastructure, while **flow.json** defines how traffic moves through that infrastructure. In our work, this setup is used to train and evaluate reinforcement learning-based multi-intersection traffic signal control systems under realistic urban traffic conditions.

2.2 Comparative summary of datasets

Table 1: General Dataset Characteristics

Dataset	Data Type	Size	What it Contains	Annotations
Performance Measurement System	Real highway sensor data	Thousands of highway sensors across California; multi-year measurements	Traffic flow, speed, occupancy, timestamps	Sensor metadata including location and lane
Manchester Traffic Dataset	Real-world traffic counts dataset	338 count points, 37,572 raw traffic records, 4,631 daily flow records, 8,419 directional flow records	Vehicle counts at road count points, traffic by vehicle class, annual average daily flow, directional traffic flow	Metadata including count location, estimation method, road type
SUMO Traffic Scenarios	Simulation dataset	Ranges from single intersection to district-level and full city simulations	Road network topology, intersections, lanes, vehicle routes, traffic signals	Vehicles and routes defined explicitly in simulation files
Hangzhou 4×4 Traffic Signal Control Dataset	Real traffic data converted into a 4×4 grid simulation environment	16 interconnected intersections; simulation duration ~3600 s	Road network (roadnet.json) and traffic demand flows (flow.json)	Used for reinforcement learning-based multi-intersection traffic signal control benchmarks
Traffic Flow Dataset (Zenodo)	Real-world networks + simulated traffic demand for SUMO	10 networks (6 synthetic + 4 real), 22 optimization instances, each with 1200 traffic demand inputs	Traffic volume inputs, generated vehicle routes, SUMO network definitions	Traffic demand definitions per network

Table 2: Inputs, Outputs, and Use cases

Dataset	Inputs	Outputs	Use Cases
Performance Measurement System	Sensor measurements at time intervals (e.g., 5-min aggregation)	Predicted traffic speed or flow in forecasting tasks	Traffic flow prediction, congestion analysis, spatio-temporal modelling
Manchester Traffic Dataset	Traffic counts from manual and automatic roadside sensors	Traffic volume statistics such as vehicles per day	Traffic demand analysis, urban traffic statistics, traffic forecasting
SUMO Traffic Scenarios	Network file (.net.xml), route file (.rou.xml), configuration (.sumocfg)	Vehicle travel time, delay, queue length, throughput	Traffic signal optimization, traffic simulation experiments, reinforcement learning training
Hangzhou 4×4 Traffic Signal Control Dataset	roadnet.json, flow.json	Average travel time used for evaluation	Reinforcement learning-based traffic signal control benchmarks
Traffic Flow Dataset (Zenodo)	Traffic volume inputs (τ) used by JTRouter to generate routes	Vehicle routes and traffic simulation outputs	Traffic signal timing optimization research

Chapter Three: Literature Reviews

3.1 Introduction

Recent research on traffic signal control has shifted from traditional fixed-time and rule-based methods to more adaptive and data-driven approaches. Early optimization techniques improved basic performance but lacked scalability in complex urban networks. With the rise of machine learning, reinforcement learning has become a key method for learning dynamic and efficient signal control policies. More recent studies focus on multi-intersection coordination using different state representations, reward functions, and learning strategies. However, differences in experimental settings and evaluation of metrics make direct comparison difficult. This section presents a comparative overview of existing works to highlight their strengths, limitations, and research gaps.

3.2 Modern technologies and frameworks used

Modern smart traffic signal systems use advanced technologies to improve traffic flow, reduce congestion, and enhance road safety. One of the most important approaches is Multi-Agent Systems (MAS), where each traffic signal acts as an intelligent agent that communicates and cooperates with nearby signals. These systems often rely on Artificial Intelligence (AI) and Machine Learning (ML) to predict traffic patterns and make real-time decisions. Reinforcement Learning (RL) is widely used to optimize signal timing dynamically based on traffic conditions. Smart traffic systems also use Internet of Things (IoT) devices such as sensors, cameras, and connected vehicles to collect live traffic data. Computer Vision techniques help detect vehicles, pedestrians, and traffic density through video analysis. Modern frameworks frequently integrate Vehicle-to-Infrastructure (V2I) communication to enable interaction between vehicles and traffic lights. Cloud computing and Edge Computing are used to process large amounts of data quickly and efficiently. Simulation tools like SUMO and MATLAB are commonly used for testing and development. These technologies together create adaptive, decentralized, and intelligent traffic management systems.

3.3 Papers Comparison

Table 1: Methodology Comparison

Paper	Publisher	Main Goal	State Representation	Action Space	Reward Design
CoLight	ACM KDD / arXiv	Network-level coordination using communication between intersections	Neighbouring traffic states using graph attention (GAT-based spatial-temporal features)	Select traffic signal phase at each intersection	Queue length + delay reduction across neighbours

PressLight	AAAI	Max-pressure RL for traffic signal control	Lane-based queue length and traffic density	Discrete phase selection	Pressure-based reward (incoming vs outgoing queues)
IPDALight	Journal of Systems Architecture	Adaptive phase duration control	Traffic intensity + queue + phase duration	Phase + duration control	Travel time minimization reward
CVLight	Transportation Research Part C	RL using connected vehicle data	CV-based delay, speed, partial observation	Phase control using CV info	Delay-based reward
GPLight	IJCAI	Large-scale coordinated MARL	Flow + topology clustering features	Group-based phase selection	Group-level efficiency reward
MOMA-DDPG	arXiv	Multi-objective RL (traffic + emissions)	Queue, delay, emission features	Continuous signal control	Multi-objective reward

Table 2: Results, Limitations, and Impact

Paper	Agent Type	Environment	Reported Results (ATT/AWT/AQL/Throughput)	Gaps	Limitations	Improvement over Fixed-Time
CoLight	Multi-agent RL (graph-based)	SUMO grid networks	Strong reduction in waiting time and higher throughput	Limited fine-grained features	High communication complexity	Strong coordination improvement
PressLight	Independent RL agents	SUMO 4×4, 6×6 grids	Reduced ATT and queue length vs baselines	No inter-agent communication	Poor scalability in large networks	Significant congestion reduction
IPDALight	Multi-agent RL	Synthetic + real scenarios	Lower travel time, faster convergence	Sensitive state design	Requires manual feature tuning	Better flow + green-wave effect
CVLight	Actor-critic agents	2×2, 5×5 + real case	Reduced delay under CV penetration	Needs connected vehicles	Limited real-world scalability	Better delay reduction
GPLight	Grouped MARL	Large-scale networks	High throughput, low delay	Grouping errors affect performance	Complex clustering	Strong scalability improvement
MOMA-DDPG	Dual-agent RL	Real datasets	Lower delay + emissions	Hard reward balancing	Sensitive tuning	Improves efficiency + environment

Table 3: Comparison of Multi-Intersection Traffic Signal Control Metrics

Method	Year	4×4 Benchmark	Metric Reported	Official Result
Fixed-Time	Baseline	Synthetic 4×4 Grid	Average Travel Time (ATT)	357–571 s
MaxPressure	2019	Synthetic 4×4 Grid	ATT	344–680 s

PressLight	2019	Synthetic 4×4 Grid	ATT	176–460 s
CoLight	2019	Synthetic 4×4 Grid	ATT	~10–20% lower ATT than PressLight depending on traffic flow
MPLight	2020	Synthetic 4×4 Grid	ATT	221–361 s

3.4 Research Gap and Proposed Innovation

Although recent smart traffic signal studies such as CoLight, PressLight, and GPLight achieved significant improvements over fixed-time traffic control, several research gaps remain unresolved. Most existing approaches focus mainly on reducing average travel time or queue length, but they often suffer limitations in state representation, scalability, communication efficiency, and reward design. For example, CoLight uses graph-based communication between agents, but its communication overhead becomes expensive in large-scale networks. PressLight lacks inter-agent communication, which reduces coordination between intersections. Other methods such as CVLight depend heavily on connected vehicle data, limiting their applicability in real-world environments with low connected vehicle penetration.

The main research gap is that current models do not effectively capture detailed real-time traffic conditions while maintaining efficient coordination among multiple agents. In addition, many studies use manually engineered state features and simple reward functions that fail to fully represent traffic dynamics such as lane occupancy, waiting patterns, and traffic imbalance.

To address these limitations, the proposed work introduces an enhanced Multi-Agent Reinforcement Learning (MARL) model based on a fine-tuned version of CoLight. The innovation focuses on two major improvements:

- **Enhanced State Representation:**
The state representation was redesigned to include richer and more informative traffic features, allowing agents to better understand congestion patterns and intersection conditions. This improves decision-making accuracy and traffic adaptability.
- **Improved Reward Function:**
A new reward mechanism was developed to balance multiple traffic objectives simultaneously, such as minimizing waiting time, reducing queue length, and improving overall traffic flow. The redesigned reward helps stabilize learning and encourages cooperative behavior among agents.

Unlike previous approaches, the proposed model aims to achieve better coordination with lower complexity while improving scalability and convergence performance. The proposed enhancements are expected to reduce average travel time, increase throughput, and provide more robust performance under dynamic traffic conditions compared to traditional fixed-time systems and baseline MARL methods.

Chapter Four: Research Methodology

4.1 Introduction

This research follows a progressive methodology for developing an intelligent traffic signal control system using Multi-Agent Reinforcement Learning (MARL). The study started with a single intersection environment to understand the behavior of reinforcement learning agents and evaluate adaptive traffic signal control under different traffic conditions. This initial phase focused on analyzing the effectiveness of state representation, reward design, and agent learning performance in a simpler and more controlled setting.

After validating the single-agent approach, the research was extended to a multi-intersection environment, where multiple agents interact and cooperate to optimize traffic flow across the network. Moving from a single intersection to multiple intersections introduced more complex challenges such as coordination between neighboring intersections, scalability, communication efficiency, and handling dynamic traffic patterns.

The proposed system is based on a fine-tuned version of CoLight. The research introduces improvements in two main areas: enhanced state representation and a redesigned reward function. The new state representation enables agents to capture more detailed traffic information, while the improved reward function helps balance traffic objectives such as reducing waiting time, minimizing congestion, and improving overall traffic flow.

The developed model was evaluated by comparing its performance with traditional fixed-time traffic signal control and baseline reinforcement learning approaches using metrics such as Average Travel Time (ATT), Average Waiting Time (AWT), queue length, and throughput. This methodology provides a systematic approach for analyzing how cooperative multi-agent learning can improve traffic signal efficiency in complex urban traffic networks.

4.2 Traditional Methods for Solving the Traffic Signal Control Problem

Before the use of Multi-Agent Reinforcement Learning (MARL), traffic signal control was mainly handled using several traditional and rule-based approaches. These methods are simpler but less adaptive to real-time traffic changes.

4.2.1 Fixed-Time Control

This is the most basic method, where traffic lights operate on a pre-defined cycle regardless of actual traffic conditions. Each phase (green, yellow, red) has a fixed duration calculated based on historical data. While easy to implement, it performs poorly under dynamic or unexpected traffic conditions.

4.2.2 Actuated Control Systems

In this approach, traffic signals respond to real-time inputs from sensors (such as loop detectors or cameras). The green time is adjusted depending on vehicle presence or queue length. However, decisions are still rule-based and limited in adaptability.

4.2.3 Coordinated (Green Wave) Systems

This method synchronizes multiple intersections along a corridor to allow continuous vehicle flow at a certain speed. Although effective for arterial roads, it struggles in complex urban networks with irregular traffic patterns.

4.2.4 Traffic Flow Optimization Models

These methods use mathematical optimization techniques such as linear programming or queueing theory to determine signal timing. They often assume simplified traffic conditions and are computationally expensive for large-scale networks.

4.2.5 Heuristic-Based Control

These systems rely on predefined rules like "if queue > threshold, increase green time." They are easy to design but cannot generalize well to unseen traffic scenarios.

4.3 Reinforcement Learning

4.3.1 Introduction

Reinforcement Learning (RL) is a branch of machine learning in which an agent learns how to make decisions by interacting with an environment. The agent improves its behavior through trial and error by selecting actions and receiving rewards or penalties based on the outcomes of those actions. The objective of reinforcement learning is to learn an optimal strategy that maximizes cumulative future rewards over time.

Unlike supervised learning, reinforcement learning does not require labeled data. Instead, learning occurs through continuous interaction with the environment and evaluation of feedback signals.

4.3.2 Core Components of Reinforcement Learning

- **Agent:** The decision-making entity responsible for selecting actions.
- **Environment:** The external system with which the agent interacts.
- **State (S):** A representation of the current condition of the environment.
- **Action (A):** A decision taken by the agent to influence the environment.
- **Reward (R):** Numerical feedback received after performing an action.
- **Policy:** The strategy used by the agent to choose actions based on states.
- **Value Function:** Estimates the expected long-term reward of states or actions.

4.3.3 Reinforcement Learning Objective

The primary objective of Reinforcement Learning (RL) is to maximize the cumulative discounted reward over time:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

Where:

- (G_t) = cumulative reward (return)
- (γ) = discount factor, where $(0 \leq \gamma \leq 1)$
- (r_{t+k+1}) = future reward received at time step $(t + k + 1)$

This objective enables the agent to learn an optimal policy that maximizes long-term performance in dynamic environments

4.4 Proposed Model Architecture

4.4.1 The development process

The development process began with building and testing a custom reinforcement learning model using the Hangzhou 1×1 dataset. This initial stage was used to validate the reinforcement learning workflow, evaluate traffic signal behavior, and analyze the interaction between states, actions, and rewards within a single-intersection environment.

After validating the approach to the 1×1 environment, the work was extended to a larger and more realistic traffic network using the Hangzhou 4×4 dataset. The final model was based on the CoLight multi-agent reinforcement learning architecture.

The selected architecture enables coordination between neighboring intersections through graph-based communication, making it suitable for large-scale urban traffic optimization. The original CoLight framework was modified in several components to improve performance and adapt the model to the Hangzhou 4×4 traffic environment. The modified

components include:

- Reward Function
- State Representation
- Graph Representation
- Discount Factor (γ)
- Training Rounds

4.4.2 State Representation

The state representation was modified to provide the agent with more informative traffic observations related to congestion and traffic flow conditions at each intersection.

The updated state design improves the model's ability to understand dynamic traffic patterns and make more effective signal control decisions.

4.4.3 Graph Representation

The graph representation component was adjusted to improve communication between neighboring intersections within the 4×4 traffic network.

This modification enhances spatial coordination and allows intersections to exchange more relevant traffic information during the learning process.

4.4.4 Reward Function

The reward function was redesigned to better reflect traffic optimization objectives such as:

- reducing vehicle waiting time,
- minimizing congestion,
- improving traffic movement efficiency.

The modified reward structure provides more stable learning behavior and improves convergence during training.

4.4.5 Loss Function

The model uses a **Mean Squared Error (MSE)** based on Temporal Difference loss.

It minimizes the squared difference between predicted and target Q-values.

The predicted Q-value represents the model's estimate of action quality.

The target Q-value is computed from reward plus discounted future reward.

The reward captures immediate improvement in traffic conditions.

Future value estimation is obtained using a target network for stability.

MSE is applied to measure and penalize prediction errors.

This allows the model to progressively improve its traffic signal control policy.

4.4.6 Training Algorithms

The training process is based on:

- Deep Q-Learning (DQN),
- Multi-Agent Reinforcement Learning (MARL),
- and Graph Attention Networks (GAT) are used within the CoLight framework.

The model also utilizes:

- Experience Replay,
- Target Network Stabilization,
- Epsilon-Greedy Exploration during training.

These methods improve training stability, exploration efficiency, and coordination between intersections in the 4×4 traffic environment.

4.5 Project Pipeline

The system is a modular reinforcement learning framework designed for multi-intersection traffic signal control. It operates as an iterative closed-loop pipeline in which experience collection, model optimization, and evaluation are repeatedly executed to progressively improve policy performance in a traffic simulation environment.

4.5.1 System Initialization (runexp.py)

The entry point of the system is responsible for configuring and initializing the entire experiment.

Functions:

- Loads configuration files for the environment, model architecture, and training parameters
- Defines directory structures for logs, datasets, and model checkpoints
- Initializes experiment settings required for execution

This module serves as the setup layer that ensures all components are correctly configured prior to training.

4.5.2 Pipeline Controller (pipeline.py)

This module acts as the central orchestration unit of the framework.

Functions:

- Manages iterative training cycles (training rounds)
- Coordinates interaction between system modules, including data generation, model updating, and evaluation
- Controls execution flow and synchronizes simulation and learning phases

It functions as the main controller of the reinforcement learning loop.

4.5.3 Experience Generation (generator.py)

This component interacts directly with the traffic simulation environment to collect training data.

Functions:

- Executes the traffic simulation using the current policy
- Controls agents at each intersection
- Collects transition data in the form of:
 - State (traffic conditions)
 - Action (signal phase selection)
 - Reward (environmental feedback)
 - Next state (resulting traffic conditions)

This module is responsible for generating the experience dataset used for learning.

4.5.4 Data Processing and Storage

Collected simulation data is stored and processed into structured training samples.

Functions:

- Stores raw interaction data in log formats (pkl, .csv)
- Converts experience tuples into structured datasets using preprocessing utilities
- Prepares training samples suitable for neural network input

This stage bridges the gap between simulation output and model training input.

4.5.5 Model Training (updater.py)

This module performs the optimization of the reinforcement learning model.

Functions:

- Loads processed training samples
- Trains the Graph Attention Network (GAT)-based CoLight model
- Updates network parameters using reinforcement learning objectives
- Optimizes the policy based on observed rewards

This component represents the learning mechanism of the system.

4.5.6 Graph Attention Network (GAT) Module

The core learning model is based on a Graph Attention Network architecture.

Functions:

- Encodes spatial and relational traffic state representations
- Models' dependencies between adjacent intersections
- Computes Q-values for traffic signal actions

The output consists of updated model parameters stored as checkpoint files.

4.5.7 Evaluation Module (`model_test.py`)

This module evaluates the performance of trained models.

Functions:

- Runs the trained policy in the simulation environment
- Computes traffic performance metrics such as delay, waiting time, and throughput
- Compares performance across different model versions

It serves as the performance assessment stage of the pipeline.

4.5.8 Model Management

The system includes mechanisms for storing and selecting optimal models.

Functions:

- Saves model checkpoints during training
- Retains best-performing models based on evaluation metrics
- Manages versioning of learned policies

This ensures that only the most effective policy is preserved.

4.5.9 Overall Pipeline Structure

1. System initialization
2. Experience generation through simulation
3. Data preprocessing and storage
4. Model training via the GAT-based learner
5. Performance evaluation
6. Model selection and checkpointing
7. Iteration of the cycle

4.6 The Used Datasets

The Hangzhou 1×1 dataset represents a single-intersection traffic scenario used for initial validation of the CoLight framework. It provides a simplified environment where the model learns basic signal control policies without inter-intersection dependencies. The dataset contains traffic flow states, signal phase actions, and corresponding rewards collected from a single junction under varying demand patterns. It is primarily used to verify convergence behavior and baseline performance of the reinforcement learning model.

The Hangzhou 4×4 dataset extends this setup to a grid of sixteen interconnected intersections, introducing spatial dependencies and traffic coordination challenges. Each intersection is modeled as a node in a graph, where traffic conditions are influenced by neighboring intersections. The dataset captures complex multi-agent interactions, including state transitions, coordinated signal actions, and network-wide reward signals. It is used to evaluate the scalability and effectiveness of CoLight’s graph-based attention mechanism in large-scale urban traffic networks.

4.7 Research Methodology Summary

The proposed research methodology is built on a deep reinforcement learning framework for traffic signal control, implemented through the CoLight architecture. It follows an iterative closed-loop process in which traffic simulations generate experience data, which is then processed and used to train a Graph Attention Network–based policy model. The trained model is periodically evaluated using traffic performance metrics such as delay, waiting time, and throughput. Based on evaluation results, the best-performing models are saved and used for subsequent training rounds. The methodology is validated on both single-intersection (Hangzhou 1×1) and multi-intersection (Hangzhou 4×4) datasets to demonstrate scalability from simple to complex urban traffic networks.

Chapter Five: Research Results

In this chapter, the tools, frameworks, and technologies used in developing and implementing the proposed traffic signal control system are presented. The chapter also describes the software environment, reinforcement learning frameworks, simulation tools, and programming technologies used throughout the research.

5.1 Tools and Technologies Used

The proposed traffic signal control system was developed and evaluated using several software tools, frameworks, and technologies related to reinforcement learning, traffic simulation, and deep learning. The main tools and technologies used in this research are as follows:

- **CoLight Framework:**
The research is based on the CoLight multi-agent reinforcement learning framework for network-level traffic signal control. The framework utilizes Graph Attention Networks (GAT) to enable communication and coordination between neighboring traffic intersections.
- **Python Programming Language:**
Python was used as the primary programming language for implementing reinforcement learning models, data preprocessing, training pipeline, and evaluation procedures.
- **Deep Reinforcement Learning (DRL):**
The system uses Deep Q-Learning (DQN) and Multi-Agent Reinforcement Learning (MARL) techniques for adaptive traffic signal optimization.
- **Graph Attention Networks (GAT):**
Graph-based neural network architectures were used to model spatial relationships and communication between multiple intersections within the traffic network.
- **Traffic Simulation Environment:**
The experiments were conducted using the Hangzhou traffic simulation datasets integrated within the CoLight environment to simulate realistic urban traffic conditions.
- **JSON-based Dataset Files:**
The traffic environment configuration and traffic flow generation were managed using:
 - roadnet.json for road network topology and intersection definitions
 - flow.json for vehicle generation and traffic demand patterns
- **Experience Replay and Target Networks:**
These reinforcement learning techniques were used to improve learning stability and convergence during training.
- **Docker Environment:**
Docker was used to simplify environment setup and ensure compatibility between required dependencies and the traffic simulation framework.

- **GitHub Repository:**

The implementation and modified CoLight framework were managed using GitHub for source code organization and version control.

5.2 Evaluation Methods

Several evaluation methods and performance metrics were used to measure the effectiveness and robustness of the proposed enhanced CoLight model. The evaluation process focused on comparing the modified model with the original CoLight implementation under identical experimental conditions.

The evaluation methods used in this research are summarized as follows:

- **Multiple Random Seed Evaluation:**

To ensure reliable and unbiased evaluation, five different random seed values were used. Five separate experiments were conducted, with each experiment trained using a different random seed value.

- **Training Rounds:**

Each experiment was trained for 30 rounds to evaluate the convergence behavior and performance stability of the reinforcement learning model.

- **Mean and Standard Deviation Analysis:**

The mean and standard deviation of the experimental results were computed to analyze the consistency, robustness, and statistical stability of the proposed enhancements.

- **Comparative Performance Evaluation:**

The enhanced CoLight model was directly compared with the original CoLight implementation without modifications using the same datasets, random seeds, and training configurations.

- **Traffic Performance Metrics:**

The evaluation focused on several important traffic performance indicators, including:

- Average Travel Time (ATT)
- Average Waiting Time (AWT)
- Queue Length
- Traffic Throughput
- Congestion Reduction

- **Simulation-Based Evaluation:**

The model was evaluated within a realistic multi-intersection traffic simulation environment based on the Hangzhou 4×4 traffic dataset to analyze performance under dynamic urban traffic conditions.

- **Training Stability Evaluation:**

The effectiveness of the modified reward function, state representation, and clipping factor was evaluated based on convergence stability, learning efficiency, and overall traffic optimization performance.

5.3 Research Results

The experimental results showed that our model achieved an **Average Travel Time (ATT) of 422.24 seconds**, demonstrating efficient traffic movement across the network. The model also achieved a traffic **throughput of 2790.43 vehicles**, indicating effective traffic flow optimization and improved vehicle movement between interconnected intersections. In terms of traffic delay reduction, the model achieved an **Average Waiting Time (AWT) of 220.71 seconds**, reflecting the ability of the enhanced reinforcement learning model to reduce vehicle waiting times and improve traffic signal decision-making.

The Average Queue Length (AQL) achieved by the model was 7.86, showing stable congestion management and balanced traffic control performance across the intersections. In addition, **the number of unfinished vehicles was reduced to 134.12 vehicles**, indicating improved traffic clearance efficiency and better overall network performance.

To evaluate the consistency and robustness of the proposed model, statistical analysis was performed using the **standard deviation** of the experimental results obtained from the **five random seed experiments**. The model achieved a **travel time standard deviation of 55.99**, a **throughput standard deviation of 111.33**, and a **waiting time standard deviation of 147.74**.

Furthermore, **the queue length standard deviation was 2.59**, while **the unfinished vehicle standard deviation reached 51.47**. These results demonstrate that our model maintained stable and consistent performance across different experimental runs and traffic conditions. Overall, the research results demonstrate the effectiveness of the proposed modifications in improving traffic signal control performance, reducing congestion, enhancing traffic flow efficiency, and maintaining stable reinforcement learning behavior in multi-intersection traffic environments.

5.4 Experiments Results Comparison

The model was evaluated using five experiments with different random seed values (0, 7, 42, 123, and 999). Each experiment was trained for 30 rounds using the Hangzhou 4×4 real-world traffic dataset. The final evaluation results were computed using the mean and standard deviation across all experiments to ensure reliable and unbiased performance analysis

Table 1: Results Comparison

Experiment	Random Seed	Final Average Travel Time	Final Throughput	Final Average Waiting Time	Final Average Queue Length	Final Unfinished Vehicles
Experiment 1	0	310.44	2920	58.39	2.77	63
Experiment 2	7	313.48	2926	56.65	3.01	57
Experiment 3	42	317.26	2927	53.38	3.16	56
Experiment 4	123	328.39	2917	62.00	3.68	66

Experiment 5	999	326.42	2919	51.84	3.62	64
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Table 2: Overall Results Summary

Metric	Mean	Standard Deviation
Average Travel Time	319.19	7.90
Throughput	2921.8	4.44
Average Waiting Time	56.45	4.04
Average Queue Length	3.25	0.39
Unfinished Vehicles	61.2	4.44

5.6 Our Model vs. CoLight Results Comparison

This section presents a detailed comparative evaluation between the proposed enhanced CoLight model and the original CoLight model as introduced in the CoLight paper. The comparison is carried out using a set of key traffic performance indicators, along with their corresponding standard deviation values, to assess not only the effectiveness of the proposed modifications but also the stability and consistency of the learning process under different experimental conditions. The analysis highlights the impact of the improvements introduced in our model, including the modifications to the state representation, reward function, and clipping factor, in relation to the original CoLight framework.

Performance Metrics Comparison

Metric	Baseline	Enhanced	Better Result
Average Travel Time ↓	421.8	422.24	Baseline
Throughput ↑	2780.76	2790.43	Enhanced
Average Waiting Time ↓	239.54	220.71	Enhanced
Average Queue Length ↓	7.86	7.86	Nearly Equal
Unfinished Vehicles ↓	142.33	134.12	Enhanced

Standard Deviation Comparison

Metric	Baseline	Enhanced	Better Result
Travel Time Std ↓	54.12	55.99	Baseline
Throughput Std ↓	113.72	111.33	Enhanced
Waiting Time Std ↓	164.17	147.74	Enhanced
Queue Length Std ↓	2.65	2.59	Enhanced
Unfinished Vehicles Std ↓	58.27	51.47	Enhanced

Chapter Six: Conclusion

The study presented an enhanced version of the CoLight framework for multi-intersection traffic signal control. The proposed modifications focused on improving state representation, reward design, and training stability through adjustments to the clipping factor. A rigorous experimental setup was adopted to ensure a fair comparison with the original CoLight model as described in the literature. The evaluation was conducted using multiple random seeds to assess robustness and generalization under different stochastic conditions. The results demonstrated that the proposed model achieves more efficient traffic management in terms of key performance indicators such as delay reduction and improved traffic flow. In addition, the model showed improved consistency across repeated experiments, reflecting better training stability. While some metrics showed marginal differences compared to the baseline, the overall behavior indicated more balanced network-level performance. The comparative analysis confirmed that the introduced enhancements contribute positively to multi-agent coordination in traffic environments. Furthermore, the findings highlight the importance of refined state and reward design in reinforcement learning-based traffic control. Overall, the proposed approach provides a more stable and effective extension of the original CoLight methodology for real-world traffic scenarios.

6.1 Key Findings

This section summarizes the main findings derived from the experimental evaluation of the proposed enhanced CoLight model. The results are based on multiple training runs using different random seeds and a direct comparison with the original CoLight framework. The findings highlight the overall impact of the proposed modifications on traffic signal control performance.

- The enhanced CoLight model improves traffic signal control performance compared to the original CoLight framework.
- Modifying state representation allows the model to capture traffic conditions more effectively.
- The redesigned reward function contributes to better learning behavior and improved traffic coordination.
- Adjusting the clipping factor helps stabilize the training process across different experiments.
- The model demonstrates consistent performance when evaluated using multiple random seeds.
- Statistical analysis shows improved robustness and reliability of the proposed approach.
- The system achieves better overall traffic efficiency, particularly in reducing congestion-related metrics.

- The improvements lead to more stable convergence during training compared to the baseline model.
- Some metrics remain close to the original model, indicating balanced but controlled changes in behavior.
- Overall, the proposed modifications provide a meaningful enhancement to the CoLight methodology.

6.2 Challenges and Future Directions

During the development and evaluation of the proposed enhanced CoLight model, several practical and methodological limitations were observed that are important to acknowledge. In addition, these limitations naturally lead to several promising research directions that can further improve the system's performance and applicability in real-world traffic environments.

- One of the main challenges is the high variability in traffic dynamics, which makes learning stable and optimal policies difficult across all scenarios.
- Training reinforcement learning models require significant computational resources and time, especially in multi-intersection environments.
- The performance of the model is sensitive to hyperparameter tuning, particularly the reward structure and exploration settings.
- Coordination between multiple agents introduces complexity in learning efficient global policies while maintaining local performance.
- Despite improvements, some traffic scenarios still show limited gains, indicating room for further optimization.
- The reliance on simulated environments may limit direct generalization to fully real-world deployments.
- Future work can focus on improving the generalization of different urban layouts and unseen traffic patterns.
- Integrating additional real-time data sources such as connected vehicles and IoT sensors could enhance decision-making accuracy.
- More advanced reinforcement learning techniques, such as hierarchical or transformer-based approaches, could further improve scalability.
- Finally, extending the system to fully deployable smart city traffic management platforms represent a promising future direction.

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